

Recap

Proof by cases

We can prove q if we can prove

$$p_1 \rightarrow q \quad p_2 \rightarrow q$$

$$p_3 \rightarrow q$$

$$p_n \rightarrow q$$

(the above are cases)

and $(p_1 \vee p_2 \dots \vee p_n)$

This is usually accomplished implicitly by noting p_n is the “otherwise case”

In terms of rules of inference, this is based on the logical equivalence

$$(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q) \equiv (p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$$

Weve seen some cases of this, but we’ll deal with it properly in **chapter 5**

Next week

We will learn that $1^2 + 2^2 + 3^2 + 4^2 \dots + n^2 = \frac{n((n+1)(2n+1))}{6}$

A consequence is that $n(n+1)(2n+1)$ is always divisible by 6 when n is a natural number,

In fact this remains true when n is any integer.

One Piece

Lets use case analysis to show $n(n+1)(2n+1)$ is always divisible by 3,

Thought process

If n is a multiple of 3, this is easy, so part of the problem is done.

We will apply a proof by cases

(Looking ahead to set notation, the remainder is in $\{0, 1, 2\}$)

It follows $n = 3k + r$ for some integer k and some $r \in \{0, 1, 2\}$

Case 1

Thus $n = 3k$ so $n(n+1)(2n+1) = 3[k(n+1)(2n+1)]$

- Replace the first n with $3k$
- Factor out the 3 in the second part before []

Which is a multiple of 3,

Case 2

The remainder is 1

Thus

$$n = 3k + 1 \text{ so } 2n + 1 = 2(3k + 1) + 1 = 6k + 3 = 3(2k + 1)$$

$$\text{and so } n(n+1)(2n+1) = 3[n(n+1)(2n+1)]$$

which is a multiple of 3

Case 3

Otherwise the remainder is 2

In this case, $n = 3k + 2$

$$\text{so } n + 1 = (3k + 2) + 1 = 3(k + 1)$$

$$\text{and } n(n+1)(2n+1) = 3[n(n+1)(2n+1)]$$

which is again a multiple of 3

Conclusion

Since in every case $n(n+1)(2n+1)$ is a multiple of 3, and the cases are exhaustive,

$$n(n+1)(2n+1)$$

Is unconditionally a multiple of 3.

Our argument form loaded like:

1. $p_1 \rightarrow q$
2. $p_2 \rightarrow q$
3. $p_3 \rightarrow q$
- _____
4. $\therefore (p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge (p_3 \rightarrow q)$ (conjunction of 1. and 2. and then 3.)
5. $\therefore (p_1 \vee p_2 \vee p_3) \rightarrow q$ (logically equiv to 4.)
6. $p_1 \vee p_2 \vee p_3$ (preamble)
- _____
7. $\therefore q$ (Motus Ponnens of 5. and 6.)

Where p_1 is "The remainder when n is divided by 3 is 0"

p_2 is "the remainder is 1"

p_3 is "the remainder is 2"

q is " $n(n+1)(2n+1)$ is a multiple of 3"

Putting it all together

1. $n(n+1)(2n+1)$ is a multiple of 3 (Example result)
 2. $n(n+1)(2n+1)$ is even (By exercise)
 3. $\exists k n(n+1)(2n+1) = 3k$
 4. $\exists k (n(n+1)(2n+1)) = 2k$
 5. $n(n+1)(2n+1) = 3l$ (Existential instantiation of 3.)
 6. $n(n+1)(2n+1) = 2m$ (Existential instantiation of 4.)
 7. $3l = 2m$ (Subs of equals from 5. and 6.)
- $3l$ is the product
 - $2m$ is even
8. 3 is even OR l is even (result from last week)
 9. 3 is not even (Definition of even)
 10. l is even (Disjunctive Syllogism of 8. and 9.)
 11. $\exists k (l = 2k)$ (Definition of even on 10.)
 12. $l = 2o$ (Existential instantiation)
 13. $n(n+1)(2n+1) = 3(2o) = 6o$

Exercise

(will be on assignment 3)

Use case analysis to prove $\forall n [n(n+1)(2n+1)]$ is even

Note

if n is divided by 3 (using long division) the remainder must be less than 3.

Existence by cases

Goal: Lets try to show that

$$\exists x \exists y (x \text{ is irrational and } y \text{ is irrational and } x^y \text{ is rational})$$

Easy case

Maybe the $\sqrt{2}^{\sqrt{2}}$ is rational

If this is the case then:

$$x = \sqrt{2} \text{ and } y = \sqrt{2}$$

is our example

Case 2

Otherwise $\sqrt{2}^{\sqrt{2}}$ is irrational

and we know 2 irrational numbers, maybe our example is:

$$\sqrt{2}^{(\sqrt{2}^{\sqrt{2}})}$$

or:

$$\left(\sqrt{2}^{\sqrt{2}}\right)^{(\sqrt{2}^{\sqrt{2}})}$$

or

$$\left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}}$$

if we borrow from calc 2 that $(a^b)^c = a^{bc}$

Aside:

We know $a^{\frac{p}{q}}$ is $\sqrt[q]{a * a * a * a * \dots * a}$

• p factors

but what does $a^{\sqrt{2}}$ even mean?

Calculus says $e^{\sqrt{2} \ln(a)}$ makes sense.

The example $x = \sqrt{2}^{\sqrt{2}}$ and $y = \sqrt{2}$

then works since

$$x^y = \left(\sqrt{2}^{\sqrt{2}}\right)^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} * \sqrt{2}} = \sqrt{2}^2 = 2 = \frac{2}{1}$$

$\therefore$ In both cases we have an example,

Chapter 2

- Sets (2.1 & 2.2)
- Functions (2.3)
- Sequences (2.4)
- Sums (2.4)
- Cardinalities (instead of Matrices) (2.5)